

Model-Free Predictive Current Control of PMSM Drives Based on Extended State Observer Using Ultralocal Model

Yongchang Zhang , Senior Member, IEEE, Jialin Jin, and Lanlan Huang

Abstract—Conventional model predictive current control (MPCC) is a powerful control strategy for three-phase inverters that has the advantages of simple concept, quick response, easy implementation, and good performance. However, MPCC is sensitive to machine parameter variation, and the performance degrades substantially if a mismatch exists between the model parameters and real machine parameters. Model-free predictive current control (MFPC) based on an ultralocal model, which uses only the input and output of the system without considering any motor parameters, has been proposed to solve this problem in this article. Since parameters are not required, the robustness of the control system is improved. However, conventional MFPC based on an ultralocal model uses many control parameters, which increases the tuning work. Furthermore, the control performance is not ideal at low sampling frequency. This article proposes an improved MFPC based on the extended state observer of PMSM drives that does not require motor parameters and needs less tuning work and lower computational time while achieving the better performance in terms of current harmonics, tracking error, and dynamic overshoot. The proposed method is compared to conventional MPCC and MFPC, and the effectiveness is confirmed by the simulation and experimental results.

Index Terms—Current control, extended state observer (ESO), predictive control, robustness, ultralocal model.

I. INTRODUCTION

IN RECENT years, permanent magnet synchronous motor (PMSM) drives have received widespread attention in industry applications and electric vehicles because of the merits of high efficiency and high-power density. In PMSM drives, the stator current directly affects the dynamic and steady-state performance of the motor drive system [1].

Manuscript received August 9, 2019; revised December 20, 2019; accepted January 15, 2020. Date of publication February 5, 2020; date of current version October 30, 2020. This work was supported by the National Natural Science Foundation of China under Grant 51577003. (Corresponding author: Yongchang Zhang.)

Y. Zhang and J. Jin are with the Inverter Technologies Engineering Research Center of Beijing, North China University of Technology, Beijing 100144, China (e-mail: yozhang@ieee.org; 353761790@qq.com).

L. Huang was with the North China University of Technology, Beijing 100144, China. She is now with the Gui Zhou Aerospace Control Technology Corporation, LTD, Guiyang 550009, China (e-mail: hll08020123@163.com).

Color versions of one or more of the figures in this article are available online at <https://ieeexplore.ieee.org>.

Digital Object Identifier 10.1109/TIE.2020.2970660

The conventional PMSM control method is field-oriented control (FOC), which controls the stator current vector of the PMSM to control the torque of the motor directly. FOC is widely used and has a good steady-state performance. However, FOC requires the proportional-integral controller's parameters to be tuned. Direct torque control (DTC) is another conventional control method for PMSM drives that directly uses the stator flux linkage and output torque of the motor as the control target. DTC has many advantages, such as simple structure, quick dynamic response, and strong robustness, but DTC suffers from high torque ripples and variable switching frequency [2], [3].

Recently, model predictive control (MPC) has attracted increasing attention due to the advantages of the clear concept, high dynamic performance and easy implementation [4], [5]. According to the different control variables, MPC can be divided into model predictive torque control (MPTC) and model predictive current control (MPCC), which have achieved good control performance in PMSM control. MPCC is a high-performance control method for PMSM drives [5]–[9]. However, MPCC relies heavily on the accuracy of the machine parameters [10]. In reality, the machine parameters may vary with the operating point and environment; for example, temperature changes will alter the motor's stator resistance and stator inductance. If the parameter variations is not considered, the performance of MPCC deteriorates substantially [11]. For example, the steady-state and dynamic performance will decrease, the current will show a steady-state error and the noise will increase during the operation of the motor [12].

To solve the problem of machine parameter dependence, various control methods have been proposed in the literature. For example, the online parameter identification method has been proposed to obtain accurate machine parameters [10], [13], [14]. In [14], an online parameter estimation method based on a discrete-time dynamic model for PMSM is proposed. Two affine projection algorithms are designed to estimate the stator inductance, stator resistance, rotor flux linkage, and load torque. However, online parameter identification requires a considerable number of calculations, which increases the complexity of the system and the identification accuracy directly affects the control performance of the system [13].

Disturbance observer-based methods can achieve robust control against machine parameter variations by observing and compensating the disturbances caused by parameter changes [15]. In [16], a predictive control method based on the Luenberger

observer is proposed. In [17], a stator current and disturbance sliding-mode observer based on exponential reaching law is proposed. In the 1980s, Chinese scholar Jingqing Han proposed active disturbance rejection control and the extended state observer (ESO) to estimate the system state, even when the model is unknown [18]. ESO observes the total disturbance in the system by expanding it as a state variable to compensate the model uncertainty and parameter variations [18]. However, most of them is based on the mathematical model of PMSM and they still use machine parameters [19]. Furthermore, they are usually complicated in principle and require more tuning work.

Recently, the model-free control method proposed by Fliess and Join [20] has received wide attention in the fields of intelligent transportation, energy system management, and so on. Some studies of model-free predictive current control (MFPCC) based on an ultralocal model [20] for a PMSM drive system have been conducted [21]. However, this method suffers from complex tuning work and the performance may be deteriorated with relatively low sampling frequency, as confirmed in this article. In [22], another kind of MFPCC based on current difference detection is applied to PMSM. However, this control strategy requires double current sampling in one control period, which increases the calculation burden of the system [23]. Furthermore, this method requires a high sampling frequency to achieve the satisfactory performance.

To achieve robust control against parameter variations for PMSM drives, improve the steady and dynamic performance of MFPCC [21] with low sampling frequency and simplify the complex tuning work, this article proposes an MFPCC-ESO method based on an ultralocal model [20] and ESO [18]. This method uses only the information of the present and past current to predict the future current value without using any PMSM parameters. Although the model-free control using ultralocal model for a PMSM drive has been proposed in [21], there are noticeable differences between the conventional MFPCC and the proposed MFPCC with ESO. First, the control parameters to be tuned or designed in the proposed method is much less than that of conventional MFPCC. Second, the performance of the proposed method is better than that of conventional MFPCC in terms of current THD and dynamic overshoot. Finally, the computational time of the proposed method is also lower than that of conventional MFPCC. A detailed comparison of the conventional MPCC, MFPCC, and the proposed method is carried out in this article. The simulation and experimental results confirm the effectiveness of the proposed MFPCC-ESO.

II. MODEL OF THE PMSM DRIVE SYSTEM

This article is based on a surface-mounted PMSM (SPMSM), whose dq -axis inductances are equal. To avoid synchronous transformation and reduce the complexity of the control system, the mathematical model of the PMSM is expressed in a stationary $\alpha\beta$ frame using complex vectors, as in [1]

$$\mathbf{u}_s = R_s \mathbf{i}_s + \frac{d\boldsymbol{\psi}_s}{dt} \quad (1)$$

$$\boldsymbol{\psi}_s = L_s \mathbf{i}_s + \boldsymbol{\psi}_r \quad (2)$$

where $\boldsymbol{\psi}_r = \psi_f e^{j\theta_e}$ is the rotor flux vector; $\mathbf{u}_s$, $\mathbf{i}_s$, R_s , $\boldsymbol{\psi}_s$, L_s , ψ_f , θ_e , and $\omega = d\theta_e/dt$ are the stator voltage vector, stator current vector, stator resistance, stator flux vector, synchronous inductance, permanent magnet flux, and electrical rotor angle, respectively; and ω is the electrical rotor speed.

The machine equations of the stator current can be expressed in the standard state-space form as

$$\frac{d\mathbf{i}_s}{dt} = \frac{1}{L_s} (\mathbf{u}_s - R_s \mathbf{i}_s - j\omega \boldsymbol{\psi}_r). \quad (3)$$

III. CONVENTIONAL MFPCC USING THE ULTRALOCAL MODEL

The first-order ultralocal model of a single-input single-out system is expressed as [24]

$$\dot{y} = F + \alpha u \quad (4)$$

where u and y are control and output variables, respectively; α represents a nonphysical scaling factor selected by the designer; and F represents the known and unknown part of the system.

Using the ultralocal model in (4), the control law of model-free control is given as [20]

$$u = \left(-\hat{F} + \dot{y}^* + K_p e \right) / \alpha \quad (5)$$

where $\hat{F}$ donates the estimated value of F , y^* is the desired output, K_p is the proportional gain, and $e = y^* - y$ donates the tracking error.

In [21], the principle of model-free control using ultralocal model is applied to the current control of a PMSM drive and the corresponding controller is called ‘‘conventional MFPCC.’’ The reference of stator voltage in synchronous dq reference is calculated as

$$\begin{cases} u_d^* = \frac{-\hat{F}_d + di_d^*/dt + \hat{K}_{dp} (i_d^* - i_d)}{\alpha_d} \\ u_q^* = \frac{-\hat{F}_q + di_q^*/dt + \hat{K}_{qp} (i_q^* - i_q)}{\alpha_q} \end{cases} \quad (6)$$

where the subscript ‘‘ d ’’ and ‘‘ q ’’ mean the d -axis and q -axis component in synchronous reference.

IV. PRINCIPLE OF THE PROPOSED MFPCC-ESO

A. ESO Based on Ultralocal Model

In this article, according to formulas (3) and (4), the ultralocal mathematical model of PMSM can be expressed in a stationary $\alpha\beta$ frame using a complex vector as

$$\frac{d\mathbf{i}_s}{dt} = \mathbf{F} + \alpha \mathbf{u}_s \quad (7)$$

where $\mathbf{F} = (-R_s \mathbf{i}_s - j\omega \boldsymbol{\psi}_r)/L_s$ is assumed to be the unknown part and $\alpha = 1/L_s$ is the gain of the input value.

Based on the ultralocal model of PMSM in (7), a linear ESO taking $\mathbf{i}_s$ and $\mathbf{F}$ as state variables with the feedback of current

error can be constructed as [25]

$$\begin{cases} e_{rr} = z_1 - \hat{i}_s \\ \dot{z}_1 = z_2 + \alpha \mathbf{u}_s - \beta_1 e_{rr} \\ \dot{z}_2 = -\beta_2 e_{rr} \end{cases} \quad (8)$$

where $z_1 = \hat{i}_s$ and $z_2 = \hat{F}$ are the estimation of $\mathbf{i}_s$ and F in (7); β_1 and β_2 are the error feedback gains of the observer.

B. ESO Parameter Design

In the proposed method, parameters β_1 and β_2 affect the system performance, so it is necessary to design the parameters of β_1 and β_2 . Equation (8) can be rewritten in matrix form as

$$\begin{cases} \dot{z} = Az + B\mathbf{u}_s + D(y - \hat{y}) \\ \hat{y} = Cz \end{cases} \quad (9)$$

where $z = [z_1 \ z_2]^T$, $\hat{y}$ is the observation of y , $A = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$, $B = \begin{bmatrix} \alpha \\ 0 \end{bmatrix}$, $C = [1 \ 0]$, $D = \begin{bmatrix} \beta_1 \\ \beta_2 \end{bmatrix}$.

According to (9), the characteristic equation of the ESO can be expressed as

$$|sI - (A - DC)| = s^2 + \beta_1 s + \beta_2 \quad (10)$$

where I refers to the identity matrix.

To make the roots of the characteristic equation both fall at $-\omega_0$, β_1 , and β_2 are obtained as

$$\beta_1 = 2\omega_0 \quad (11)$$

$$\beta_2 = \omega_0^2 \quad (12)$$

where ω_0 is the bandwidth of the ESO and it determines the steady-state and dynamic performance of the observer.

It is not intuitive to directly design proper value of ω_0 . In this article, the design of ω_0 will be carried out in z domain. The formula (8) can be discretized as

$$\begin{cases} e_{rr}(k) = \hat{i}_s(k) - \mathbf{i}_s(k) \\ \hat{i}_s(k+1) = \hat{i}_s(k) + T_{sc} \left(\hat{F}(k) + \alpha \mathbf{u}_s(k) \right) - \beta_{01} e_{rr}(k) \\ \hat{F}(k+1) = \hat{F}(k) - \beta_{02} e_{rr}(k) \end{cases} \quad (13)$$

where $\beta_{01} = T_{sc}\beta_1$ and $\beta_{02} = T_{sc}\beta_2$ are the gains of observer. The values of β_{01} and β_{02} affect the distribution of the closed-loop poles of the system, thereby affecting the stability of the observer. Therefore, reasonable values of β_{01} and β_{02} must be selected to ensure the stability of the system and to ensure that the controller achieves the good control performance.

According to (13), the transfer function of ESO can be obtained as

$$G(z) = \frac{\hat{i}_s(z)}{\mathbf{i}_s(z)} = \frac{\beta_{01}z - \beta_{01} + \beta_{02}t_{sc}}{(z-1)^2 + \beta_{01}z - \beta_{01} + \beta_{02}t_{sc}} \quad (14)$$

and its characteristic equation is

$$z^2 + (\beta_{01} - 2)z + 1 - \beta_{01} + \beta_{02}t_{sc} = 0. \quad (15)$$

Considering $\beta_{01} = 2\omega_0 t_{sc}$ and $\beta_{02} = \omega_0^2 t_{sc}$, the poles of $G(z)$ is solved from (15) as

$$z_{1,2} = 1 - \omega_0 t_{sc} \quad (16)$$

and ω_0 can be calculated from the z domain poles as

$$\omega_0 = \frac{1 - z_{12}}{t_{sc}}. \quad (17)$$

ω_0 is such selected that $z_{1,2}$ are within the unit circle of the z domain. Generally, if ω_0 is too small (z_{12} is close to 1), the observer's dynamic performance will be worse, and if ω_0 is too large (z_{12} is close to zero), the robustness of the system will be affected and may lead to system divergence [26]. For PMSM drives, the inner current loop usually requires a high bandwidth to satisfy the requirements of dynamic response. Hence, in this article, z_{12} is set to 0.15 to achieve quick response and the corresponding ω_0 is 8500 when the sampling frequency is 10 kHz. For other sampling frequency such as 5 and 20 kHz, the corresponding ω_0 are 4250 and 17 000, respectively. The value of α is set to 50 in various sampling frequency.

C. Reference Voltage Vector Calculation

According to (7), $\mathbf{u}_s^k$ can be obtained using first-order discretization as

$$\mathbf{u}_s^k = \frac{\mathbf{i}_s^{k+1} - \mathbf{i}_s^k}{\alpha T_{sc}} - \frac{\hat{F}(k)}{\alpha} \quad (18)$$

where F is replaced by $\hat{F}$ because F can not be directly obtained.

The final voltage reference can be obtained from (18) by assuming $\mathbf{i}_s^{k+1}$ reaches its reference value $\mathbf{i}_s^{\text{ref}}$ as

$$\mathbf{u}_s^{\text{ref}} = \frac{\mathbf{i}_s^{\text{ref}} - \mathbf{i}_s^k}{\alpha T_{sc}} - \frac{\hat{F}(k)}{\alpha} \quad (19)$$

where $\mathbf{u}_s^{\text{ref}}$ is the reference value of output voltage vector and it is synthesized by SVM to generate the final gating pulses in this article.

Considering one-step delay in digital implementation, (19) can be rewritten as

$$\mathbf{u}_s^{\text{ref}} = \frac{\mathbf{i}_s^{\text{ref}}(k+2) - \mathbf{i}_s^{k+1}}{\alpha T_{sc}} - \frac{\hat{F}(k+1)}{\alpha} \quad (20)$$

where

$$\mathbf{i}_s^{\text{ref}}(k+2) = \mathbf{i}_s^{\text{ref}}(k) \cdot e^{j(\theta_r^k + 2\omega_r T_{sc})} \quad (21)$$

$\mathbf{i}_s^{k+1}$ and $\hat{F}(k+1)$ can be obtained from (13).

V. SIMULATION AND EXPERIMENTAL RESULTS

To confirm the effectiveness of the proposed method, simulations and experimental tests are implemented with SPMSM based on a two-level inverter. The control diagram of the PMSM drive system based on the proposed method is shown in Fig. 1. Table I lists the parameters of the control system used in the digital simulation and experimental tests. The sampling frequency is set to 10 kHz unless explicitly indicated. The value of ω_0 is 4250, 8500, and 17 000 when the sampling frequency is 5, 10, and 20 kHz, respectively.

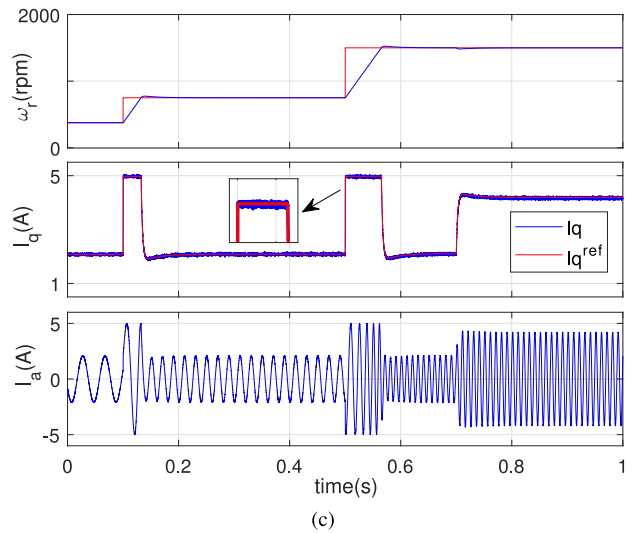
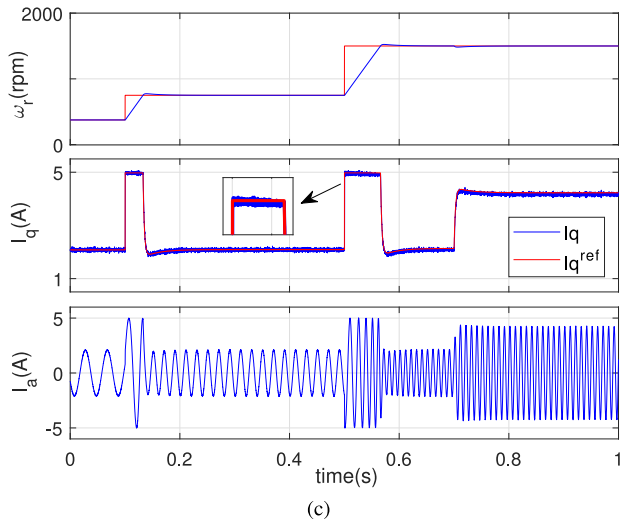
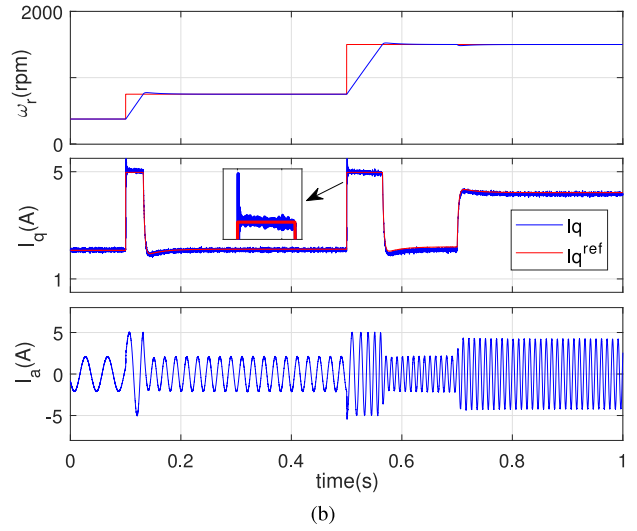
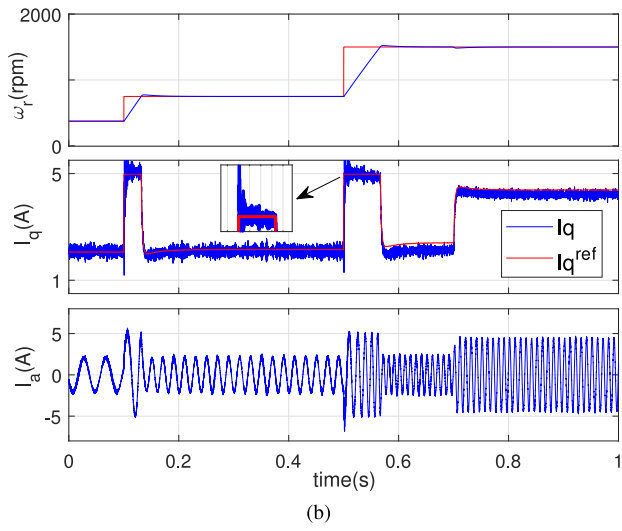
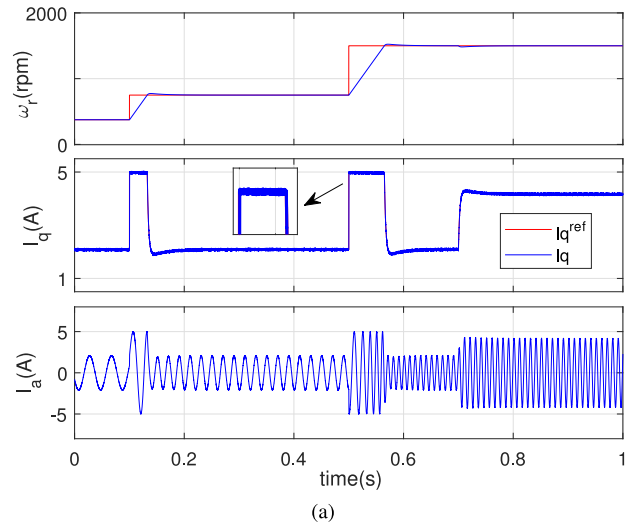
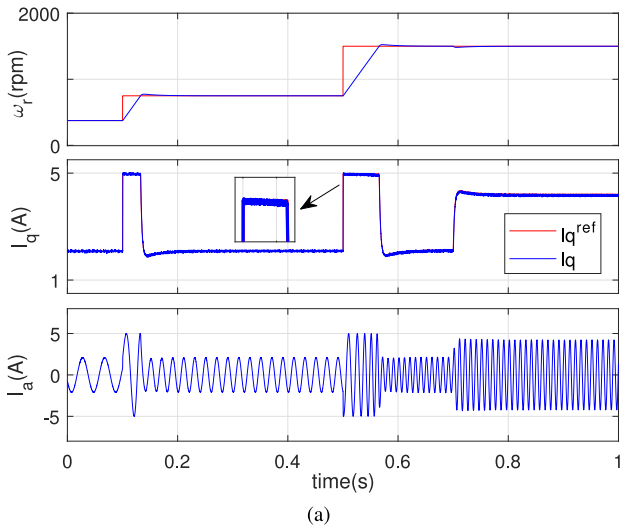


Fig. 2. Simulated dynamic response with a sampling frequency of 10 kHz under accurate parameters for (a) conventional MPCC, (b) MFPC, and (c) proposed MFPC-ESO.

Fig. 3. Simulated dynamic response with a sampling frequency of 20 kHz under accurate parameters for (a) conventional MPCC, (b) MFPC, and (c) proposed MFPC-ESO.

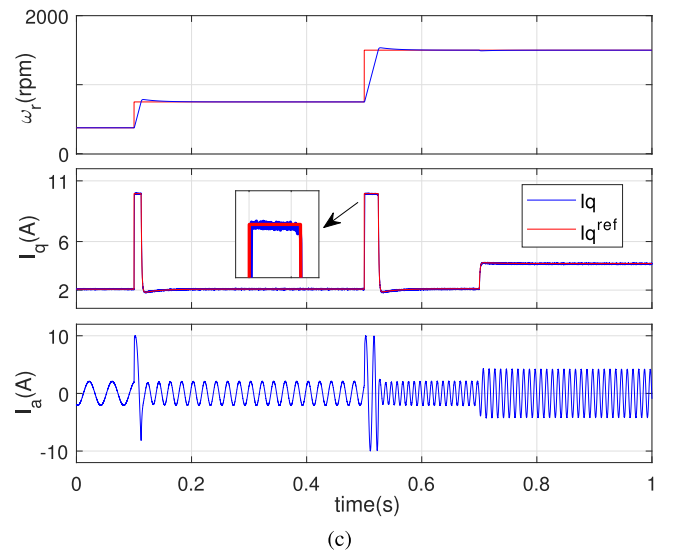
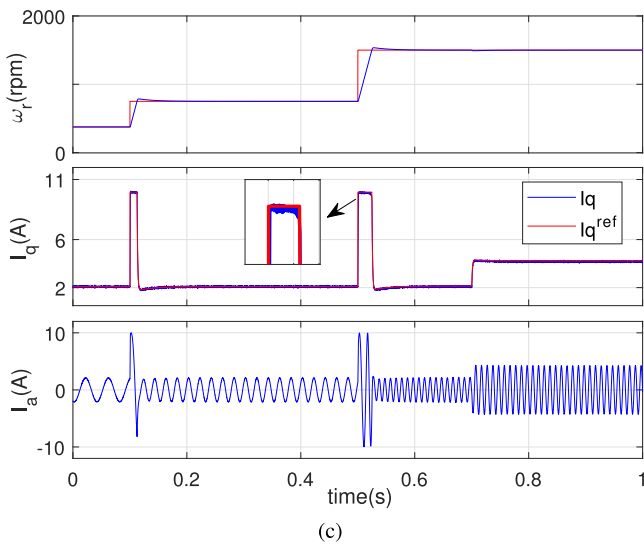
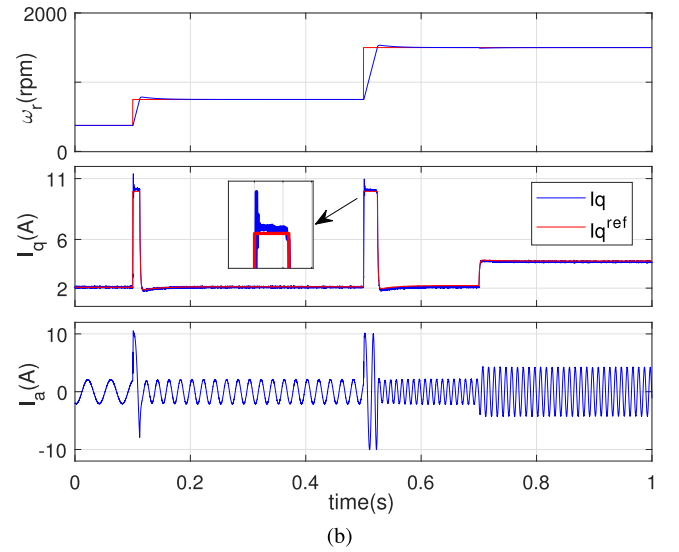
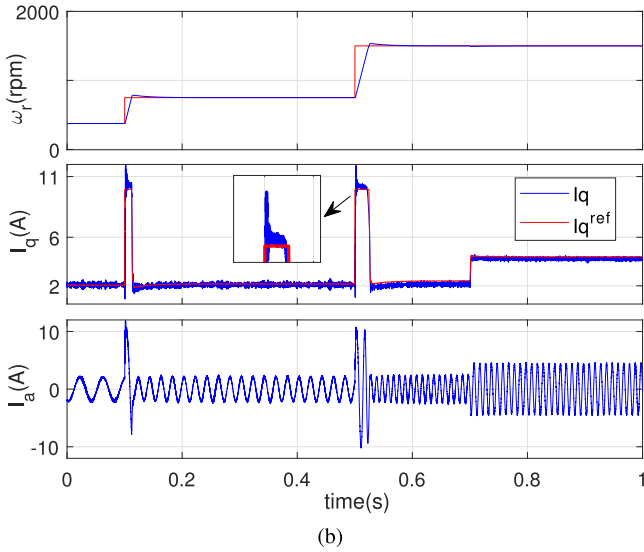
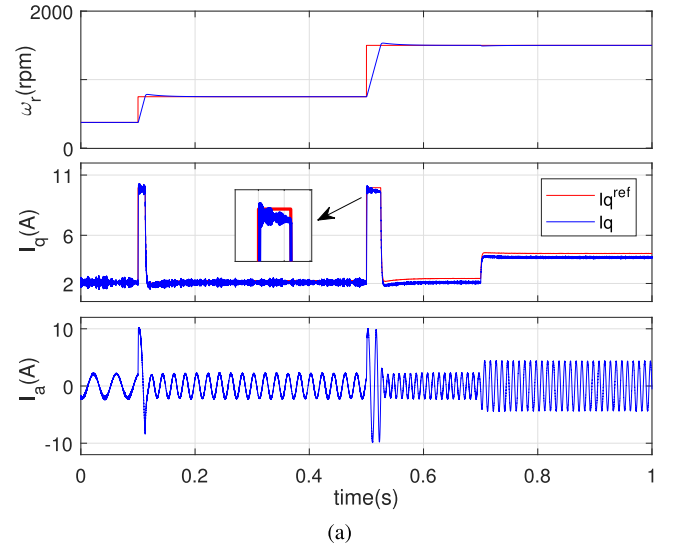
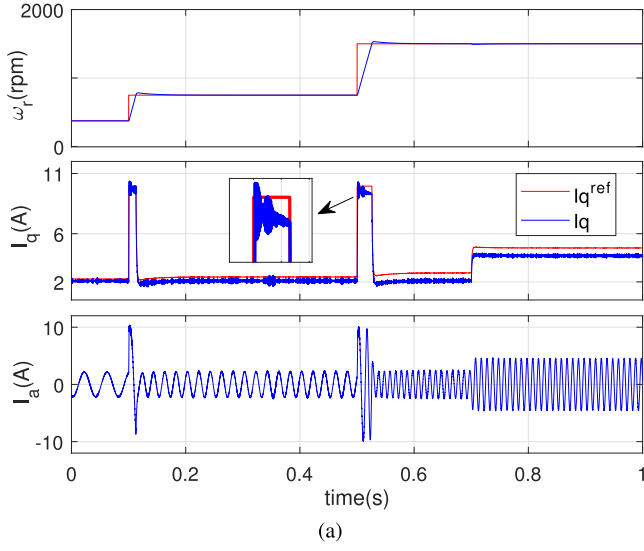


Fig. 4. Simulated dynamic response with a sampling frequency of 10 kHz under mismatched parameters for (a) conventional MPCC, (b) MFPCC, and (c) proposed MFPCC-ESO.

Fig. 5. Simulated dynamic response with a sampling frequency of 20 kHz under mismatched parameters for (a) conventional MPCC, (b) MFPCC, and (c) proposed MFPCC-ESO.

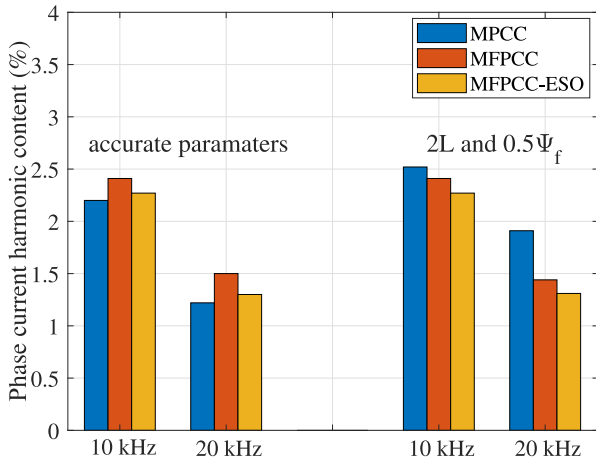


Fig. 6. Simulated FFT analysis for a phase current with accurate parameters and mismatched parameters.

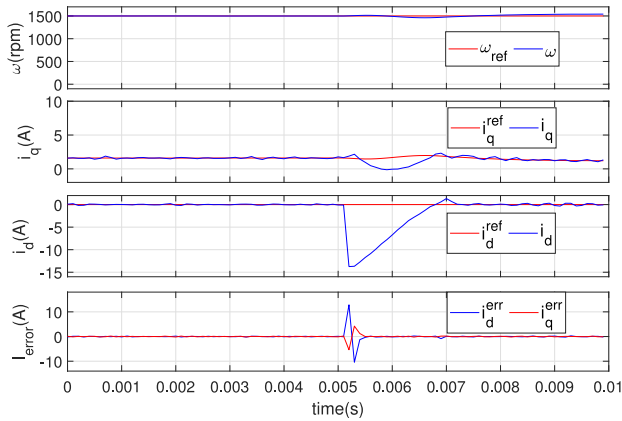


Fig. 7. Simulated response of MFPC-ESO with a sudden parameter variations under the condition of 10 kHz sampling frequency.

B. Experimental Results

The proposed MFPC-ESO, conventional MPCC and MFPC are experimentally tested on a two-level inverter-fed 2.4 kW/380 V/100 Hz SPMSM drive platform, as shown in Fig. 8. The rotor speed, the q -axis current, and its reference are displayed via an on-board DA converter, and the phase current is directly measured by a current probe. All waveforms are displayed by a digital oscilloscope.

Fig. 9 presents the current THD for conventional MPCC, MFPC, and the proposed MFPC-ESO with accurate machine parameters at rated speed of 1500 r/min. It can be seen that the proposed MFPC-ESO presents a very similar performance to conventional MPCC with accurate machine parameters. Compared to conventional MFPC, the current THDs in the proposed method are reduced by 19.0%, 5.8%, and 11.4% with the sampling frequency of 5, 10, and 20 kHz, respectively. The current THD reduction is more pronounced in the case of low sampling frequency of 5 kHz.

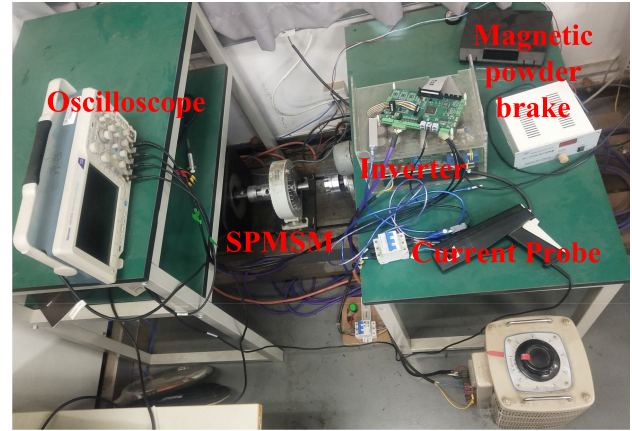


Fig. 8. Experimental platform of a two-level inverter-fed SPMSM drive.

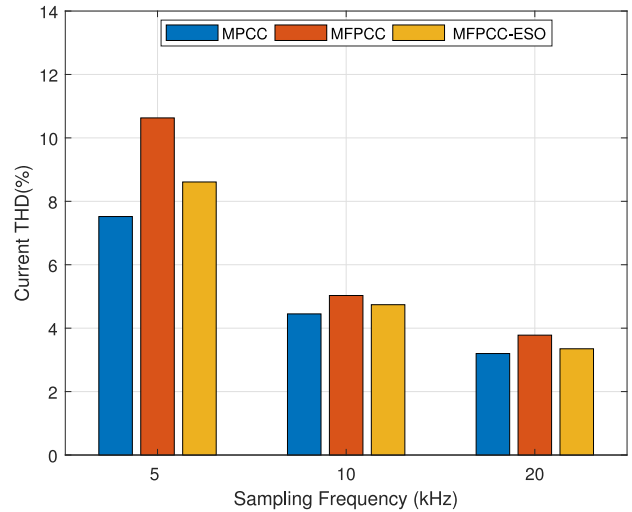
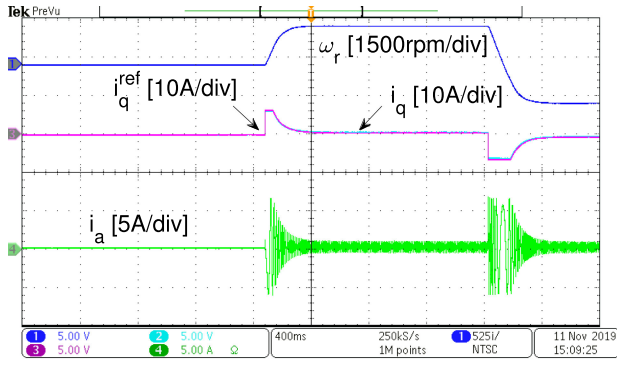


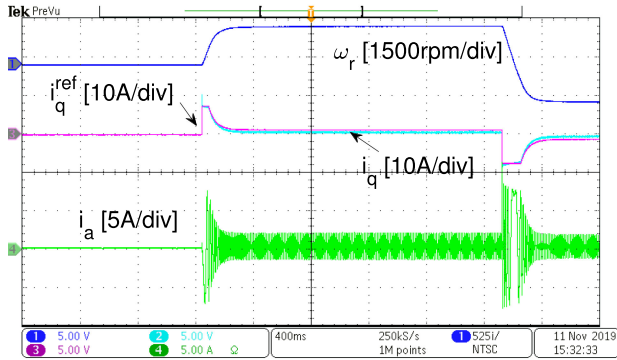
Fig. 9. Current THD of three methods under accurate parameters.

Fig. 10 presents the starting response from standstill to rated speed and high-speed reversal at rated speed for the three methods with a sampling frequency of 10 kHz under accurate parameters. It is seen that the proposed method exhibits the excellent dynamic performance, which is similar to that of conventional MPCC. For conventional MFPC, there are evident overshoot in the q -axis current during the dynamic process, which does not appear in other two methods. The experimental results are in accordance to the simulation results in Fig. 2.

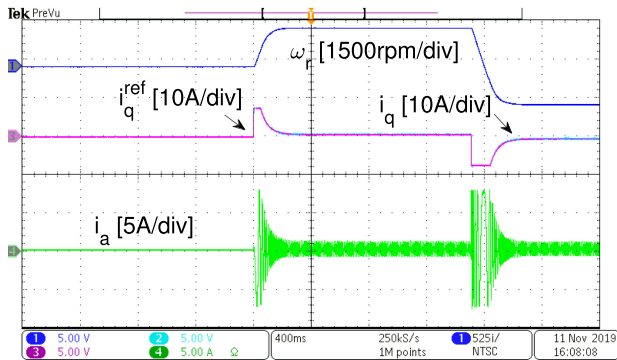
Fig. 11 shows the dynamic response to stepped load torque change with a sampling frequency of 10 kHz under accurate parameters. It is seen that the dynamic response of the three methods are very similar. The q -axis current increases quickly to balance the sudden load and the motor speed recovers to its reference value quickly. However, there are some tracking error in the conventional MFPC, which does not appear in the proposed method. The results confirm the effectiveness of the proposed method in achieving quick dynamic response with accurate machine parameters.



(a)



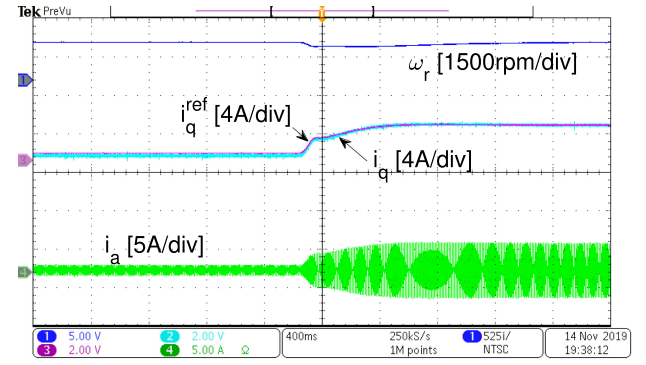
(b)



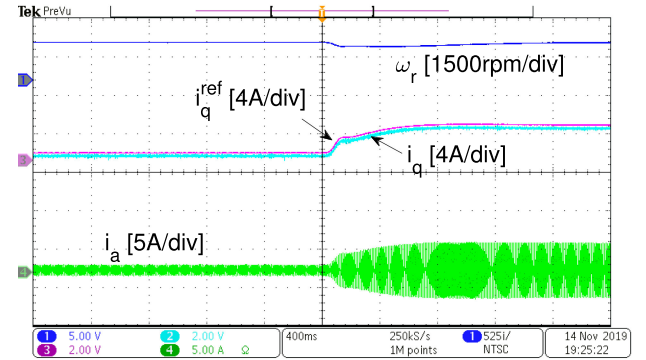
(c)

Fig. 10. Responses of starting from standstill to rated speed and high-speed reversal at rated speed with a sampling frequency of 10 kHz under accurate parameters for (a) conventional MPCC, (b) MFPCC, and (c) MFPCC-ESO (time scale, 400 ms/div).

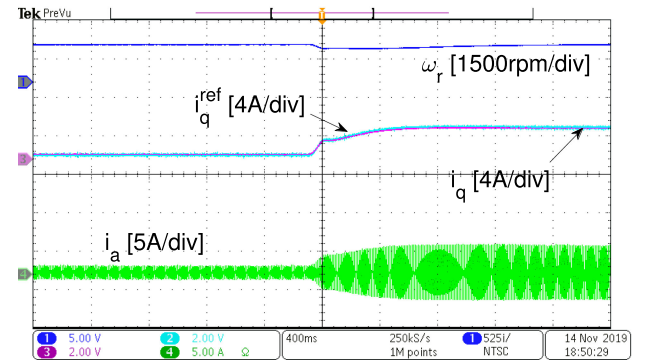
The results mentioned above are obtained with **accurate machine parameters**. When the machine parameters used in the controller differ significantly from the actual machine parameters, **the conventional MPCC even cannot start normally**. As shown in the simulation results and studied in the other literature [10], conventional MPCC suffers from the inaccurate machine parameter seriously. Considering the page limitations and that the focus of this article is to propose an improved MFPCC method based on ESO, in the following results only MFPCC and MFPCC-ESO are compared in the experiments.



(a)



(b)

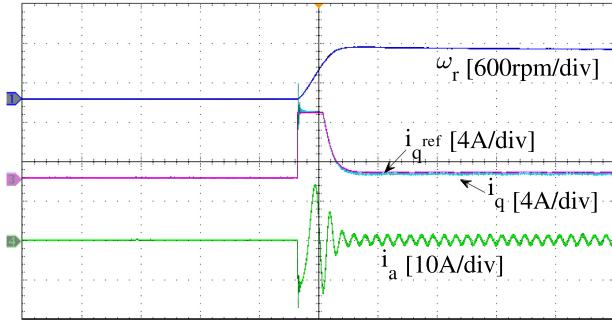


(c)

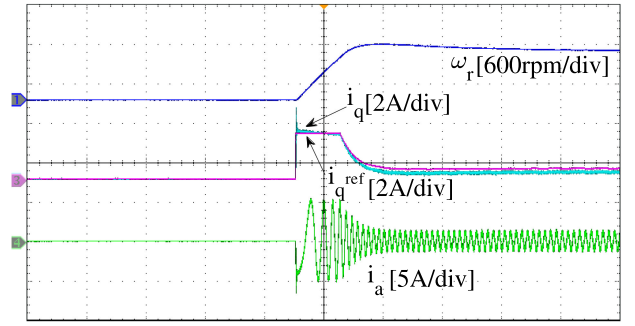
Fig. 11. Dynamic response to stepped load torque change with a sampling frequency of 10 kHz under accurate parameters for (a) conventional MPCC, (b) MFPCC, and (c) MFPCC-ESO (time scale, 400 ms/div).

Figs. 12–14 show the **dynamic responses** of MFPCC and the proposed MFPCC-ESO at **no-load** and **load with mismatched machine parameters** under the condition of 10 kHz sampling frequency. From top to bottom, the curves shown in Figs. 12–14 are the rotor speed, the reference value of the q -axis current, the actual value of the q -axis current, and the A-phase stator current.

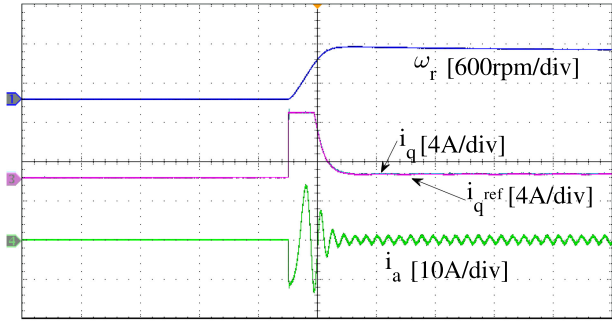
The PMSM parameters are set as follows in Fig. 12: the dq -axis inductance, stator resistance, and flux linkage in the controller are 200%, 1000%, and 70% of the actual values, respectively. In Fig. 13, the parameters are set as follows: the dq -axis inductance, stator resistance, and flux linkage in the



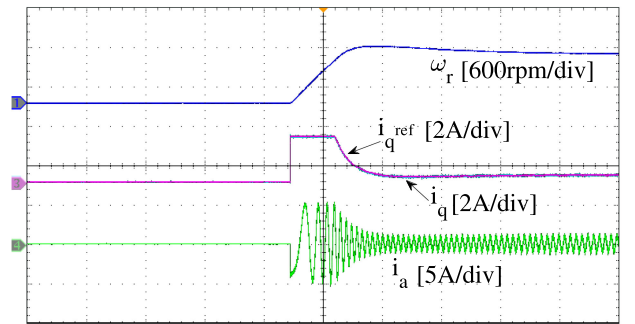
(a)



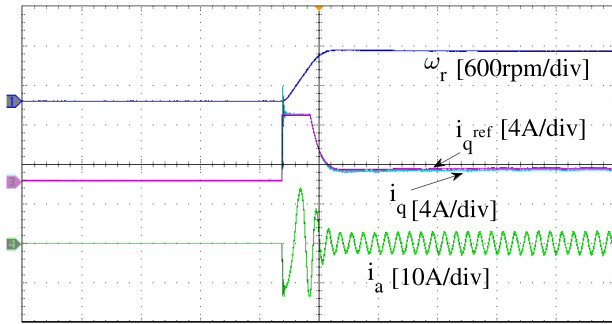
(a)



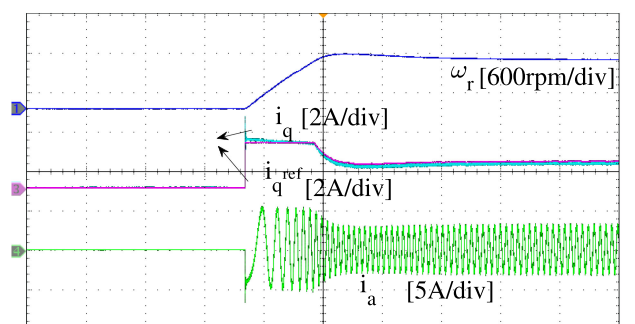
(b)



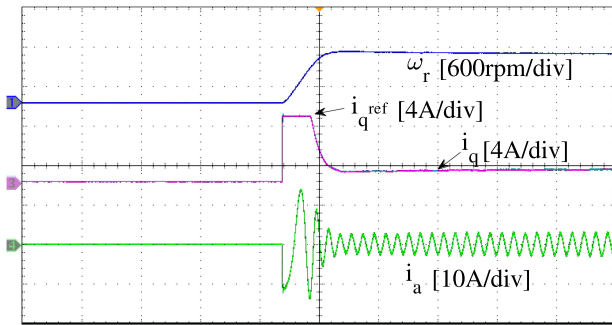
(b)



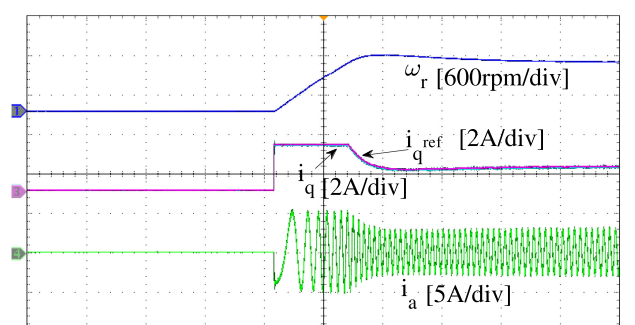
(c)



(c)



(d)



(d)

Fig. 12. Experimental results during a dynamic process with mismatched parameters at $10R$, $0.7\psi_f$, $2L$ under the condition of 10 kHz sampling frequency for (a) MFPC no-load, (b) MFPC-ESO no-load, (c) MFPC with load, and (d) MFPC-ESO with load (time scale, 100 ms/div).

Fig. 13. Experimental results during a dynamic process with mismatched parameters at $0.1R$, $2\psi_f$, $0.5L$ under the condition of 10 kHz sampling frequency for (a) MFPC no-load, (b) MFPC-ESO no-load, (c) MFPC with load, and (d) MFPC-ESO with load (time scale, 200 ms/div).

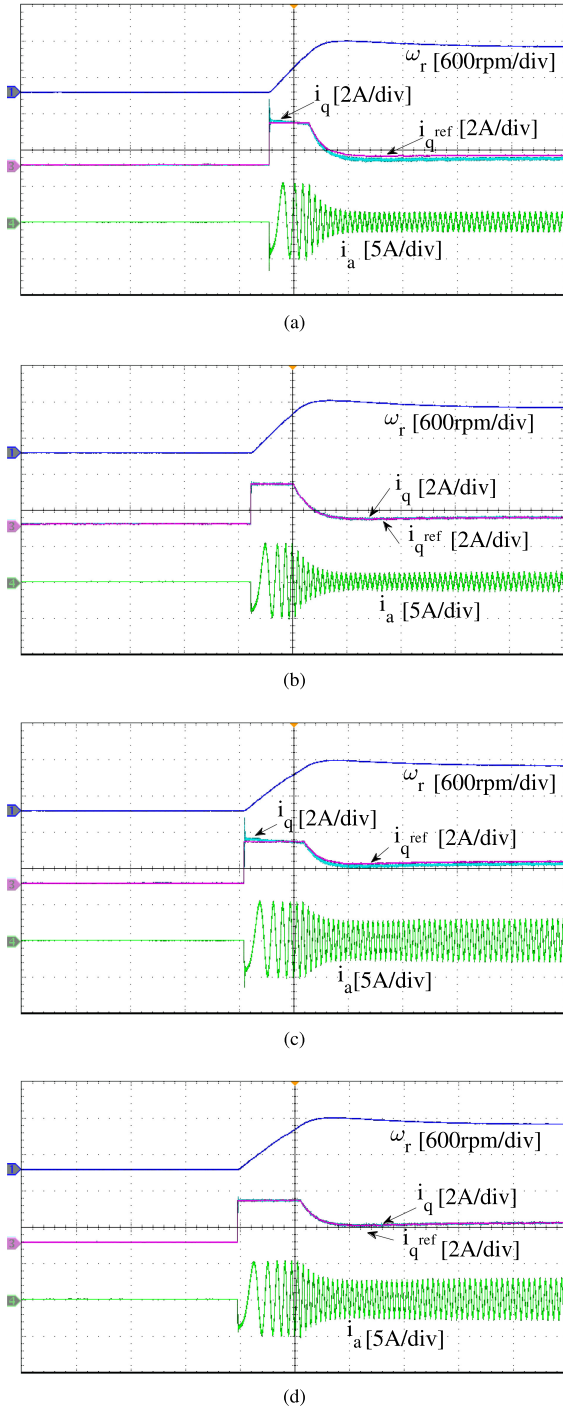


Fig. 14. Experimental results during a dynamic process with mismatched parameters at $1R$, $2\psi_f$, $2L$ under the condition of 10 kHz sampling frequency for (a) MFPCC no-load, (b) MFPCC-ESO no-load, (c) MFPCC with load, and (d) MFPCC-ESO with load. (time scale, 200 ms/div).

controller are 50%, 10%, and 200% of the actual values, respectively. In Fig. 14, the parameters are set as follows: the dq -axis inductance, stator resistance, and flux linkage in the controller are 200%, 100%, and 200% of the actual values, respectively.

The experimental results show that the proposed MFPCC-ESO has the better control performance than does MFPCC at a sampling frequency of 10 kHz. During the dynamic operation of

TABLE II
IMPLEMENTATION COMPLEXITY COMPARISON OF THREE METHODS

	conventional MPCC	MFPCC	MFPCC- ESO
Motor parameters needed	R_s , L_d , L_q , ψ_f and so on	Null	Null
Parameters to be designed or tuned	Null	K_{dp} , K_{qp} , α_d , α_q , n_F	α , ω_0
Computational time	36.97 μ s	44.76 μ s	38.89 μ s

the PMSM drive system, the current ripple of the MFPCC-ESO is smaller, the motor runs more stably, the overshoot is smaller, and the parameter robustness is stronger.

A detailed complexity comparison of the conventional MPCC, MFPCC, and the proposed method is listed in Table II. Compared to conventional MPCC, the computational time of the proposed method is only slightly increased, but the machine parameter robustness is greatly enhanced. Compared to conventional MFPCC using as much as five control parameters, the proposed method only requires to tune two parameters. Furthermore, the computational time is reduced by 13.1%. As the overall control performance of the proposed method is also better than that of conventional MFPCC, it is concluded that the proposed method outperforms MFPCC and may be a powerful alternative to conventional MPCC. In the future study, the proposed method may be extended to the speed sensorless operation.

VI. CONCLUSION

This article proposed a model-free control algorithm based on ESO using an ultralocal model for a PMSM drive system that does not require motor parameters and presented strong parameter robustness. The simulation and experimental results showed that the proposed MFPCC-ESO and conventional MPCC have a similar performance when the parameters are accurate. However, when the parameters were inaccurate, the dynamic and steady performance of the proposed method was better than that of the conventional MPCC. In contrast to the MFPCC, MFPCC-ESO needed to tune less control parameters and the computational time was also lower. Furthermore, MFPCC-ESO outperformed MFPCC in terms of current ripples, tracking error, and dynamic overshoot. The steady-state performance improvement was more pronounced with a relatively low sampling frequency of 5 kHz. The simulation and test results confirmed that the proposed method achieves good parameter robustness, fast dynamic response, and satisfactory steady-state performance.

ACKNOWLEDGMENT

The authors would like to thank H. Jiang for his support when preparing the experimental results.

REFERENCES

- [1] Y. Zhang and J. Zhu, "Direct torque control of permanent magnet synchronous motor with reduced torque ripple and commutation frequency," *IEEE Trans. Power Electron.*, vol. 26, no. 1, pp. 235–248, Jan. 2011.
- [2] L. Zhong, M. F. Rahman, W. Y. Hu, and K. W. Lim, "Analysis of direct torque control in permanent magnet synchronous motor drives," *IEEE Trans. Power Electron.*, vol. 12, no. 3, pp. 528–536, May 1997.

- [3] D. Casadei, F. Profumo, G. Serra, and A. Tani, "Foc and DTC: two viable schemes for induction motors torque control," *IEEE Trans. Power Electron.*, vol. 17, no. 5, pp. 779–787, Sep. 2002.
- [4] Y. Zhang, Y. Bai, H. Yang, and B. Zhang, "Low switching frequency model predictive control of three-level inverter-fed IM drives with speed-sensorless and field-weakening operations," *IEEE Trans. Ind. Electron.*, vol. 66, no. 6, pp. 4262–4272, Jun. 2019.
- [5] P. Cortes, M. P. Kazmierkowski, R. M. Kennel, D. E. Quevedo, and J. Rodriguez, "Predictive control in power electronics and drives," *IEEE Trans. Ind. Electron.*, vol. 55, no. 12, pp. 4312–4324, Dec. 2008.
- [6] F. Morel, X. Lin-Shi, J. Retif, B. Allard, and C. Buttay, "A comparative study of predictive current control schemes for a permanent-magnet synchronous machine drive," *IEEE Trans. Ind. Electron.*, vol. 56, no. 7, pp. 2715–2728, Jul. 2009.
- [7] H.-T. Moon, H.-S. Kim, and M.-J. Youn, "A discrete-time predictive current control for PMSM," *IEEE Trans. Power Electron.*, vol. 18, no. 1, pp. 464–472, Jan. 2003.
- [8] Y. Zhang, D. Xu, J. Liu, S. Gao, and W. Xu, "Performance improvement of model-predictive current control of permanent magnet synchronous motor drives," *IEEE Trans. Ind. Appl.*, vol. 53, no. 4, pp. 3683–3695, Jul./Aug. 2017.
- [9] Y. Zhang, D. Xu, and L. Huang, "Generalized multiple-vector-based model predictive control for PMSM drives," *IEEE Trans. Ind. Electron.*, vol. 65, no. 12, pp. 9356–9366, Dec. 2018.
- [10] X. Zhang, L. Zhang, and Y. Zhang, "Model predictive current control for PMSM drives with parameter robustness improvement," *IEEE Trans. Power Electron.*, vol. 34, no. 2, pp. 1645–1657, Feb. 2019.
- [11] H. A. Young, M. A. Perez, and J. Rodriguez, "Analysis of finite-control-set model predictive current control with model parameter mismatch in a three-phase inverter," *IEEE Trans. Ind. Electron.*, vol. 63, no. 5, pp. 3100–3107, May 2016.
- [12] J. C. Moreno, J. M. Espi Huerta, R. G. Gil, and S. A. Gonzalez, "A robust predictive current control for three-phase grid-connected inverters," *IEEE Trans. Ind. Electron.*, vol. 56, no. 6, pp. 1993–2004, Jun. 2009.
- [13] T. Boileau, N. Leboeuf, B. Nahid-Mobarakeh, and F. Meibody-Tabar, "Online identification of PMSM parameters: Parameter identifiability and estimator comparative study," *IEEE Trans. Ind. Appl.*, vol. 47, no. 4, pp. 1944–1957, Jul./Aug. 2011.
- [14] D. Q. Dang, M. S. Rafiq, H. H. Choi, and J.-W. Jung, "Online parameter estimation technique for adaptive control applications of interior PM synchronous motor drives," *IEEE Trans. Ind. Electron.*, vol. 63, no. 3, pp. 1438–1449, Mar. 2016.
- [15] J. Yang, W.-H. Chen, S. Li, L. Guo, and Y. Yan, "Disturbance/uncertainty estimation and attenuation techniques in PMSM drives—A survey," *IEEE Trans. Ind. Electron.*, vol. 64, no. 4, pp. 3273–3285, Apr. 2017.
- [16] C. Xia, M. Wang, Z. Song, and T. Liu, "Robust model predictive current control of three-phase voltage source PWM rectifier with online disturbance observation," *IEEE Trans. Ind. Informat.*, vol. 8, no. 3, pp. 459–471, Aug. 2012.
- [17] X. Zhang, B. Hou, and Y. Mei, "Deadbeat predictive current control of permanent-magnet synchronous motors with stator current and disturbance observer," *IEEE Trans. Power Electron.*, vol. 32, no. 5, pp. 3818–3834, May 2017.
- [18] J. Han, "From PID to active disturbance rejection control," *IEEE Trans. Ind. Electron.*, vol. 56, no. 3, pp. 900–906, Mar. 2009.
- [19] S. Chang, P. Chen, Y. Ting, and S. Hung, "Robust current control-based sliding mode control with simple uncertainties estimation in permanent magnet synchronous motor drive systems," *IET Electr. Power Appl.*, vol. 4, no. 6, pp. 441–450, Jul. 2010.
- [20] M. Fliess and C. Join, "Model-free control and intelligent PID controllers: Towards a possible trivialization of nonlinear control?" *IFAC Proc. Volumes*, vol. 42, no. 10, pp. 1531–1550, 2009.
- [21] Y. Zhou, H. Li, and H. Yao, "Model-free control of surface mounted PMSM drive system," in *Proc. IEEE Int. Conf. Ind. Technol.*, Mar. 2016, pp. 175–180.
- [22] C. Lin, T. Liu, J. Yu, L. Fu, and C. Hsiao, "Model-free predictive current control for interior permanent-magnet synchronous motor drives based on current difference detection technique," *IEEE Trans. Ind. Electron.*, vol. 61, no. 2, pp. 667–681, Feb. 2014.
- [23] Y. Chen, T. Liu, C. Hsiao, and C. Lin, "Implementation of adaptive inverse controller for an interior permanent magnet synchronous motor adjustable speed drive system based on predictive current control," *IET Electr. Power Appl.*, vol. 9, no. 1, pp. 60–70, 2015.
- [24] M. Fliess and C. Join, "Model-free control," *Int. J. Control*, vol. 86, no. 12, pp. 2228–2252, 2013.
- [25] Z. Gao, "Scaling and bandwidth-parameterization based controller tuning," in *Proc. Amer. Control Conf.*, vol. 6, Jun. 2003, pp. 4989–4996.
- [26] Q. Xu, M. Sun, Z. Chen, and D. Zhang, "Analysis and design of the extended state observer using internal mode control," in *Proc. 32nd Chin. Control Conf.*, Jul. 2013, pp. 5408–5413.



Yongchang Zhang (Senior Member, IEEE) received the B.S. degree from Chongqing University, Chongqing, China, in 2004, and the Ph.D. degree from Tsinghua University, Beijing, China, in 2009, both in electrical engineering.

From August 2009 to 2011, he was a Postdoctoral Fellow with the University of Technology Sydney, Ultimo, NSW, Australia. He joined the North China University of Technology, Beijing, in August 2011 as an Associate Professor. He is currently a Full Professor and the Director of

Inverter Technologies Engineering Research Center, Beijing. He has authored or coauthored more than 100 technical papers in the area of motor drives, pulsewidth modulation, and ac–dc converters. His current research interest is model predictive control for power converters and motor drives.



Jialin Jin was born in 1993. He received the B.S. degree in electrical engineering in 2017 from the North China University of Technology, Beijing, China, where he is currently working toward the M.S. degree in control science and engineering.

His research interests include model predictive control of permanent magnet synchronous motor drives.



Lanlan Huang was born in 1994. She received the B.S. degree in automation engineering and the M.S. degree in electrical engineering from the North China University of Technology, Beijing, China, in 2016 and 2019, respectively.

In September 2019, she joined Gui Zhou Aerospace Control Technology Corporation, LTD, Guiyang, China. Her research interests include model predictive control of permanent magnet synchronous motor drives.